HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2026)

Student No.: 20 Name: _____ Marks: / 39

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The stem-and-leaf diagram below shows the distribution of the numbers of driving hours of a group of taxi drivers in a week.

| Stem (tens) | Leaf (units) |
|---|---|
| 2 | 5 6 9 |
| 3 | 2 3 3 6 6 7 8 |
| 4 | 0 0 2 2 5 5 7 7 7 |
| 5 | $w$ |

The ratio of the range to the inter-quartile range of the distribution is 7 : 3.

(a) Find $w$.

(b) (i) Write down the mode of the distribution.

(ii) If two taxi drivers are randomly selected from the group, find the probability that the mean of their numbers of driving hours in the week exceeds the mode of the distribution.

(6 marks) (DSE-2026 PI Q10)

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2. It is given that $P$, $Q$ and $R$ are three points in a rectangular coordinate plane. The coordinates of $P$ and $Q$ are $(7,8)$ and $(-13,20)$ respectively. The equation of the perpendicular bisector of $PR$ is $3x+5y-27=0$. Let $C$ be the circle which passes through $P$, $Q$ and $R$.

(a) Find the equation of $C$.

(b) The tangent to $C$ at $P$ and the tangent to $C$ at $R$ intersect at the point $H$. Is $\triangle HPR$ a right-angled triangle? Explain your answer.

(6 marks) (DSE-2026 PI Q16)

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3. In Figure 2, the base of the pyramid $VABCD$ is the quadrilateral $ABCD$. The face $VAB$ of the pyramid lies on the horizontal ground. It is given that $AB = 90$ cm, $AD = 70$ cm, $CD = 100$ cm, $\angle ADC = 55°$ and $\angle BCD = 130°$.

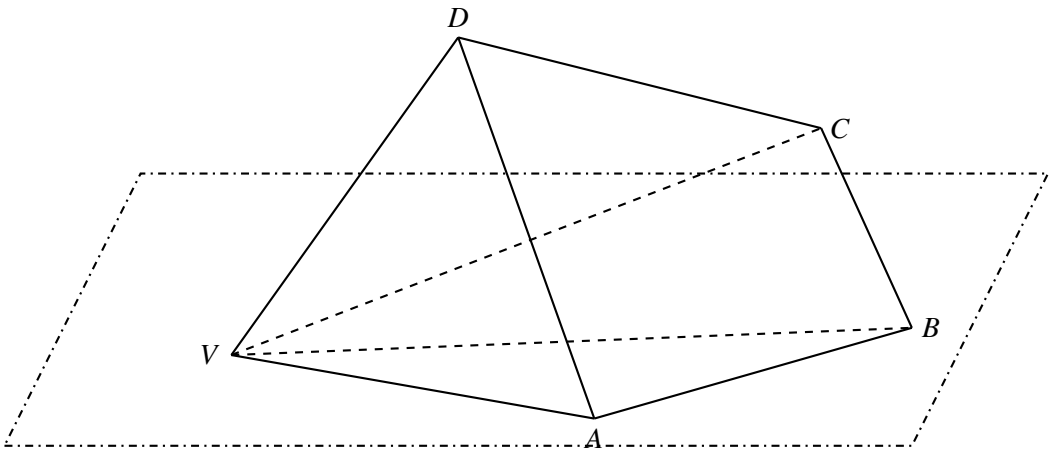



**Figure 2**

(a) Find

(i) the distance between $A$ and $C$,

(ii) $\angle BAD$.

(b) It is given that $VD = 80$ cm. A student finds that the angle between the plane $ABCD$ and the horizontal ground is 65°. The student claims that the angle between $VD$ and the horizontal ground is less than 50°. Is the claim correct? Explain your answer.

(6 marks) (DSE-2026 PI Q17)

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4. Let $G(n)$ be the $n$th term of a geometric sequence. It is given that $G(2) = 12$ and $G(5) = 768$.

(a) Find $G(1)$.

(b) Define $T(n) = \log_2 G(n)$ for all positive integers $n$. Find

(i) $T(n+1) - T(n)$,

(ii) the greatest value of $k$ such that $T(1) + T(2) + T(3) + \cdots + T(k) < 10^6$.

(7 marks) (DSE-2026 PI Q18)

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5. Let $h(x) = -x^2 + 30kx + 4x - 225k^2 - 40k + 3$, where $k$ is a positive constant. On the same rectangular coordinate system, denote the vertex of the graph of $y = h(x)$ and the vertex of the graph of $y = 14 - h(4-x)$ by $A$ and $B$ respectively.

(a) Using the method of completing the square, express, in terms of $k$, the coordinates of $A$.

(b) Write down, in terms of $k$, the coordinates of $B$.

(c) The coordinates of the point $C$ are $(2 + 48k,\ 7 - 36k)$. Let $I$ and $J$ be the in-centre and the circumcentre of $\triangle ABC$ respectively.

(i) Let $M$ be the mid-point of $AB$. Describe the geometric relationship between $CM$ and $AB$. Explain your answer.

(ii) Express, in terms of $k$, the distance between $I$ and $J$.

(iii) The coordinates of the point $D$ are $(6 - 22k,\ 6 - 15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer.

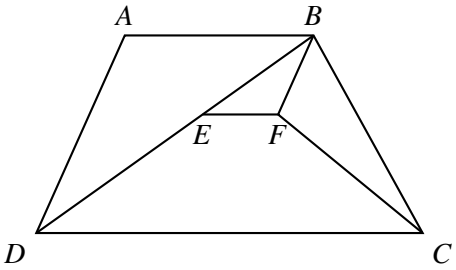
(12 marks) (DSE-2026 PI Q19)

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6. In the figure, $ABCD$ is a trapezium with $AB//DC$ and $AB : CD = 3 : 5$. Let $E$ be a point lying on $BD$ such that $BE : ED = 2 : 3$. It is given that $AB//EF$ and $AD//BF$. If the area of $\triangle ABD$ is 225 $cm^2$, then the area of the trapezium $CDEF$ is




A. 222 $cm^2$.

B. 260 $cm^2$.

C. 279 $cm^2$.

D. 285 $cm^2$.

(DSE-2026 PII Q18)

I can finish _____ questions

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